

TIME v4 Hook Implementation

Booking intent guard + telemetry for a time-credit marketplace

Uniswap v4 validates and observes booking intent during swap
BookingManager owns service rights, inventory, quote rep
protection, settlement, and lifecycle.

Hook role

beforeSwap guard
afterSwap telemetry
ZERO_DELTA

Booking role

inventory + slots
quote validity
burn/lock TIME

Frontend flow

quote → hookData → swap
then bookWithCredits

Current implementation at a glance

Grounded in the current contracts, frontend services, and docs.

TimePoolHook.sol

- Enables only beforeSwap + afterSwap
- Validates pool/router allowlists
- Decodes booking hookData
- Delegates quote/inventory validity
- Emits TimeSwapObserved

BookingManager.sol

- Provider inventory + pause
- EIP-712 quote validation
- Replay protection via usedQuotes
- Slot lock via slotTaken
- bookWithCredits burns TIME

Frontend + v4 SDK

- Quote API returns BookingQuote
- buildHookData ABI-encodes quote
- Universal Router executes V4_SWAP
- UI calls bookWithCredits after swap

Evidence: TimePoolHook.sol, BookingManager.sol, uniswapV4Service.ts, BookingCheckout.tsx

Hook permission profile

The hook is intentionally narrow: no custom accounting and no booking lifecycle mutation.

beforeSwap

ENABLED
Guard + validation

afterSwap

ENABLED
Telemetry only

Return deltas

DISABLED
No custom swap amounts

Liquidity callbacks

DISABLED
No LP lifecycle

beforeSwap / afterSwap: On

Validate booking-aware intent and emit observability.

Return deltas: Off

Avoids custom-delta/NoOp risk and keeps hook out of AMM accounting.

Liquidity callbacks: Off

Keeps LP accounting outside the marketplace hook.

Booking settlement: Not a hook permission

Settlement lives in BookingManager after final TIME ownership exists.

Activity flow: swap then book

A valid booking-aware swap does not automatically create a booking.

1. Select provider + slot

Buyer chooses service inventory

2. Get real quote

EIP-712 quote; mock is UI-only

3. Encode hookData

Frontend ABI-encodes quote

4. V4 swap executes

Universal Router -> PoolManager

5. Hook validates intent

beforeSwap checks quote + inventory

6. Telemetry emitted

afterSwap emits event only

7. Book with credits

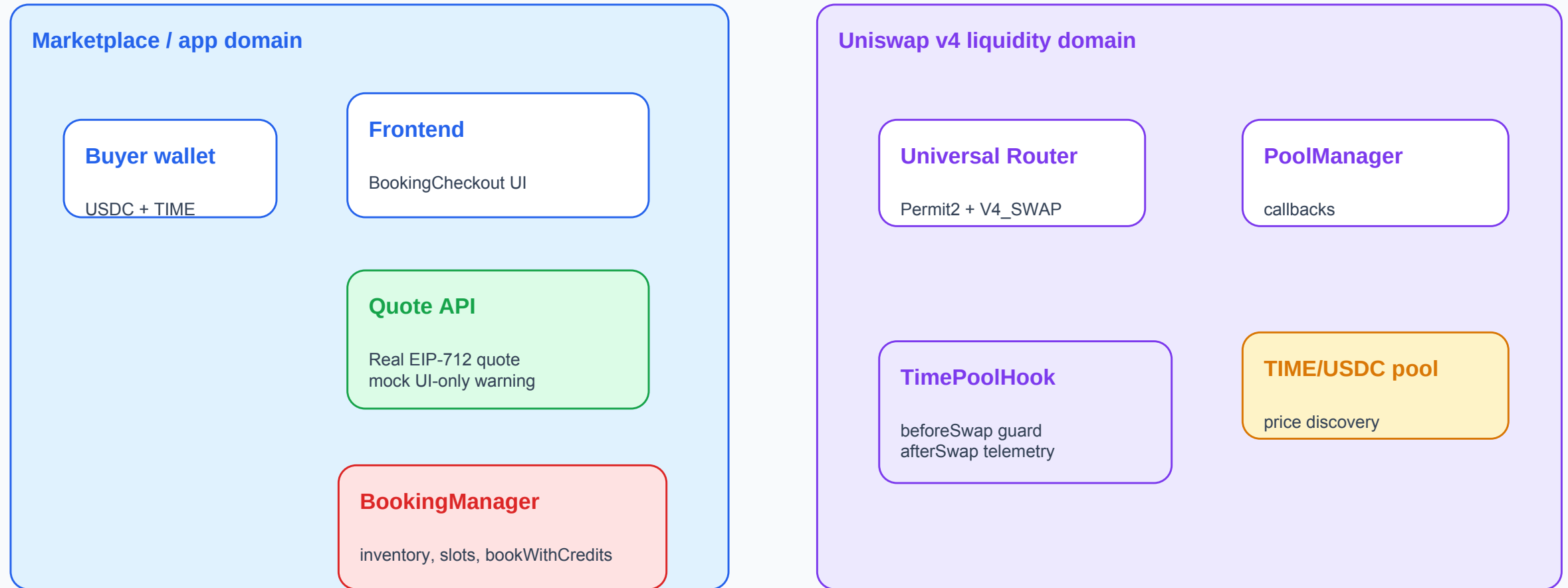
BookingManager burns TIME + locks slot

Failure mode

Swap OK + booking fail => retry
bookWithCredits with settled TIME.

System architecture

Marketplace/app domain stays separate from Uniswap v4 liquidity domain.



Invariant: booking state is created only by BookingManager after settled TIME credits are available.

Security boundary and invariants

Do not imply atomic service booking inside a v4 callback.

No booking in callbacks

Hook never calls `bookWithCredits`, burns `TIME`, or reserves slots.

Pool + router allowlists

Both callbacks call `_validatePoolAndRouter`.

hookData policy

Empty `hookData` can pass unless strict; mock is UI/dev only.

Quote replay is split

Hook `consumedHookQuote` is not `BookingManager.usedQuotes`.

afterSwap telemetry only

It emits `TimeSwapObserved` and returns zero delta.

Booking can fail after swap

UI supports retry with settled `TIME`.

Verification evidence in the repo

Existing tests and docs support the artifact claims.

Contract tests

TimePoolHook.t.sol covers permissions, owner config, allowlists, malformed hookData, buyer/hours validation, inventory bounds, invalid quote, quote replay, untrusted routers, and afterSwap telemetry.

Source files

TimePoolHook.sol shows beforeSwap/afterSwap only and ZERO_DELTA. BookingManager.sol shows usedQuotes, inventory decrement, slotTaken, TIME burn, and booking creation.

Docs + app flow

Architecture and security docs record the boundary. BookingCheckout runs swap first, then submitBooking.

Simple explanation for demos

Use this framing when walking non-specialists through the system.

1. TIME is liquid

Users can acquire fungible TIME credits through a Uniswap v4 TIME/USDC pool.

2. Hook checks intent

The hook can reject unsafe booking-aware swaps but does not create service rights.

3. Booking is separate

BookingManager burns/locks settled TIME and creates the actual booking record.

Deliverables:

- docs/diagrams/time-v4-booking-activity.excalidraw
- docs/diagrams/time-v4-system-architecture.excalidraw
- docs/presentations/time-v4-hook-implementation.pptx
- docs/presentations/time-v4-hook-implementation.pdf